

Control Flow

Bit by Bit Coding - Term 2 | Taster 2

Key Concepts

- Comparison operators compare values and return a bool: ==, !=, <, >, <=, >=.
- Logical operators combine conditions: and (both true), or (either true), not (flips).
- if / elif / else lets your program make decisions.
- Indentation (4 spaces) marks which lines belong to a block.
- Conditions can be nested for more complex logic.

Syntax Reference

Comparison & Logical Operators

```
print(5 == 5)          # True
print(5 != 3)          # True
print(10 > 20)         # False
has_key = True
has_torch = False
print(has_key and has_torch)  # False
print(has_key or has_torch)   # True
print(not has_torch)         # True
```

if / elif / else

```
score = 82
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
else:
    print("Grade: C")
```

Nested Conditionals

```
age = 16
has_id = True
if age >= 18:
    if has_id:
        print("Entry allowed")
    else:
        print("Need ID")
else:
    print("Too young")
```

Common Patterns

- Check a range with a chained comparison: if 0 <= score <= 100.
- Combine conditions: if age >= 13 and age <= 19.
- elif means "else if" - only checked when earlier branches were False.

Tips & Common Mistakes

- Use == to compare, NOT = (a single = assigns a value).
- Don't forget the colon : at the end of if / elif / else.
- Indentation matters - every line in a block must line up.
- Use the whole words and / or / not - not && or ||.

- elif is one word, not "else if".